

Priya Natarajan

SUMMARY

Machine learning engineer with 6 years of experience taking LLM and recommendation systems from research to production. Comfortable across training, serving and MLOps.

EXPERIENCE

Senior Machine Learning Engineer — Acme Robotics

Austin, TX | Mar 2021 – Present

- Fine-tuned Llama-3-8B with LoRA on 40k support transcripts, cutting escalation rate 18%.
- Built a RAG pipeline over 2M internal documents using pgvector and LangChain; p95 latency 380 ms.
- Served models with vLLM on Kubernetes, autoscaled with KEDA; handled 250 rps at peak.
- Set up drift detection with Evidently and Grafana alerting for 12 production models.

Machine Learning Engineer — Northwind Analytics

Remote | Jun 2018 – Feb 2021

- Trained gradient-boosted churn models in LightGBM on Spark-prepared features, lifting retention 4%.
- Tracked 300+ experiments in MLflow and automated retraining with Airflow DAGs.
- Deployed models as FastAPI services in Docker with CI/CD via GitHub Actions.

EDUCATION

M.S. Computer Science — University of Texas at Austin, 2018

B.Tech Computer Science — NIT Trichy, 2016

SKILLS

Languages: Python, SQL, Go

ML: PyTorch, scikit-learn, XGBoost, Hugging Face Transformers, PEFT

LLM: RAG, vector search, prompt engineering, DPO

Infra: Docker, Kubernetes, AWS SageMaker, MLflow, Airflow, Spark

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PROJECTS

Open-source reranker for hybrid retrieval (BM25 + dense)

- Cross-encoder reranker that improved recall@10 by 11 points on BEIR subsets.

CERTIFICATIONS

AWS Certified Machine Learning – Specialty, 2022